

MoodGarden — Build Tasks

Ordered implementation roadmap, milestones, and deliverables for AI Mood & Habit Journal

Implementation Rule

Work through these milestones in sequence. Each milestone must leave the application in a fully runnable and verifiable state before advancing.

PROJECT

MoodGarden (Next.js + Supabase + Hugging Face)

TOTAL MILESTONES

10 Milestones (Scaffold to Real Usage)

DOCUMENT TYPE

Milestone Execution Checklist

DEPLOYMENT TARGET

Vercel (Production) + Supabase Cloud

Milestones Overview & Build Checklist

Milestone 1 — Project Scaffold

Foundation

- ☐ `create-next-app` with TypeScript + Tailwind CSS + App Router
- ☐ Install core dependencies: `@supabase/supabase-js`, `@supabase/ssr`, `framer-motion`
- ☐ Set up `.env.local` and commit `.env.local.example` (see SETUP.md)
- ☐ Apply design tokens from DESIGN_SYSTEM.md to `tailwind.config.ts` and `globals.css` (warm off-white, sage accent, dark mode via `class`)
- ☐ Configure Google Fonts in `layout.tsx`: Fraunces (serif headings) + Inter (clean body text)

Milestone 2 — Database & Security

Supabase Postgres

- ☐ Create Supabase project and obtain project credentials (URL, anon key, service role key)
- ☐ Execute SQL schema from DATABASE_SCHEMA.md in order: tables → seed habits → RLS enable → policies → indexes
- ☐ Verify Row Level Security (RLS) isolation by testing queries with and without an active session token

Milestone 3 — Authentication

Auth / Sessions

- ☐ `lib/supabaseClient.ts` — initialize browser client
- ☐ `lib/supabaseServer.ts` — server client helper using `@supabase/ssr` for route handlers & server components
- ☐ `/login` page: email/password form calling `supabase.auth.signInWithPassword`
- ☐ `/signup` page: account registration calling `supabase.auth.signUp`
- ☐ `middleware.ts`: route guard redirecting unauthenticated traffic from `/today` and `/garden` to `/login`
- ☐ Logout action / button clearing session and redirecting cleanly

Milestone 4 — Entry Flow (Baseline)

Core Journaling

- ☐ `components/MoodPicker.tsx`: 5 emoji-style mood states (1–5 scale)
- ☐ `components/HabitChecklist.tsx`: interactive habit selectors fetched from `habits` table
- ☐ `components/EntryForm.tsx`: integrated form with journal textarea (max 500 chars), mood picker, and checklist
- ☐ `POST /api/entries` route: atomic upsert into `entries` + habit log synchronization
- ☐ `GET /api/entries/:date` route: fetch specific date entry
- ☐ Wire `/today` page: load existing entry for today (if already saved) and handle reactive updates

Milestone 5 — Garden Visualization

Interactive UI

- ☐ `lib/moodColor.ts` : maps mood score (1–5) to serene color tokens (muted clay → soft green)
- ☐ `GET /api/entries?month=YYYY-MM` route: query full month of entries for active user
- ☐ `components/GardenGrid.tsx` : 7-column calendar month grid with responsive layout
- ☐ `components/GardenTile.tsx` : day tile with mood color, bloom icon scaled by habit count, and Framer Motion stagger animation
- ☐ `components/DayModal.tsx` : click tile → smooth modal reveal displaying that day's entry, mood, habits, and sentiment
- ☐ Calendar month navigation controls (previous/next month buttons with animated transition)

Milestone 6 — AI: Sentiment Scoring

Hugging Face API

- ☐ `lib/sentiment.ts` : pure function calling DistilBERT SST-2 model on Hugging Face Inference API
- ☐ `POST /api/sentiment` route: background handler calculating score and updating `entries.sentiment_score`
- ☐ Decouple entry submission from AI latency: fire-and-forget or non-blocking async trigger
- ☐ Graceful degradation: if API times out (>5s) or fails, log null without interrupting user experience

Milestone 7 — AI: Correlation Insight

Statistical Engine

- ☐ `lib/correlation.ts` : unit-testable statistics engine evaluating mood delta with/without each habit (with `MIN_SAMPLE = 7` guardrail)
- ☐ `GET /api/insight` route: executes analysis over last 30 entries and formats natural language insight
- ☐ `components/InsightCard.tsx` : clean card displaying positive correlation or motivational message if more data is needed
- ☐ Position `InsightCard` seamlessly on `/garden` view with caching

Milestone 8 — Polish & Accessibility

UX Refinement

- ☐ `components/ThemeToggle.tsx` : smooth dark mode switch updating CSS root classes
- ☐ Mobile responsiveness audit: verify complete layout stability at 375px viewport
- ☐ Motion safety: respect `prefers-reduced-motion` in Framer Motion configurations
- ☐ Empty states: welcoming onboarding message for new accounts with 0 entries
- ☐ Loading skeletons: graceful placeholder tiles while fetching month entries
- ☐ Resilient error handling: toast notifications on failure with zero loss of unsaved text

Milestone 9 — Deployment & Verification

Production

- ☐ Push to GitHub repository and connect Vercel project with auto-deploy on main branch
- ☐ Configure 4 production environment variables in Vercel project settings
- ☐ End-to-end verification: execute full journaling and garden flow on real mobile hardware
- ☐ Re-enable Supabase email confirmation if relaxed during local testing
- ☐ Production documentation: README with architecture highlights, setup steps, and demo GIF

Milestone 10 — Real Usage & Demonstration

Portfolio Milestone

- ☐ Onboard 15–20 real users to journal daily for 14+ days
- ☐ Collect qualitative feedback and testimonials
- ☐ Document metrics (active users, entries recorded, correlation discoveries) for portfolio resume impact